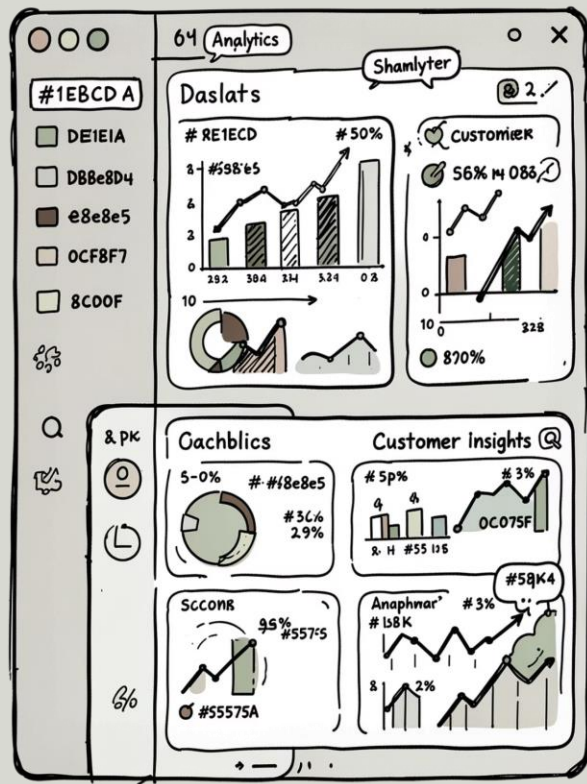




CUSTOMER SHOPPING BEHAVIOR ANALYSIS

A comprehensive data-driven study uncovering spending patterns, customer segments, product preferences, and return reasons – using SQL analysis and Power BI visualization across 4,390 rows of transactional data.

DATA ANALYTICS

SQL + POWER BI

BEAUTY AND
SKINCARE

PROJECT OVERVIEW

GOAL

Analyze customer shopping behavior using data from multiple tables to uncover insights into spending patterns, customer segments, product preferences, and reasons behind product returns – guiding strategic business decisions.

7

TABLES

39

COLUMNS

4,390

ROWS

DATASET AT A GLANCE

- **7 tables:** Customers, Order Items, Orders, Products, Returns, Reviews
- **39 columns** covering demographics, transactions, products & feedback
- **4,390 rows** of transactional and behavioral data

DATASET STRUCTURE

The analysis draws from six interconnected tables spanning the full customer journey – from acquisition to purchase to review and return.

CUSTOMERS

customer_id, name, city, state, gender, age_group, signup_date, acquisition_channel

ORDERS

order_id, order_date, order_status, payment_method, sales_channel, gross_amount, discount_amount, shipping_fee, final_amount, delivered_date

ORDER ITEMS

order_item_id, product_id, quantity, unit_price, discount_pct, item_total

PRODUCTS

product_name, category, concern, skin_type, key_ingredient, size, mrp, cost_price, stock_qty, launch_date

RETURNS

return_id, return_date, return_reason, refund_status

REVIEWS

review_id, rating, review_date



AGE GROUP & GENDER INSIGHTS

25-34 DOMINATES

The 25-34 age group leads with **632 orders (50.56%)** and the highest total discount usage at **₹62,428**, making them the most active and deal-responsive segment.

18-24 ACTIVE

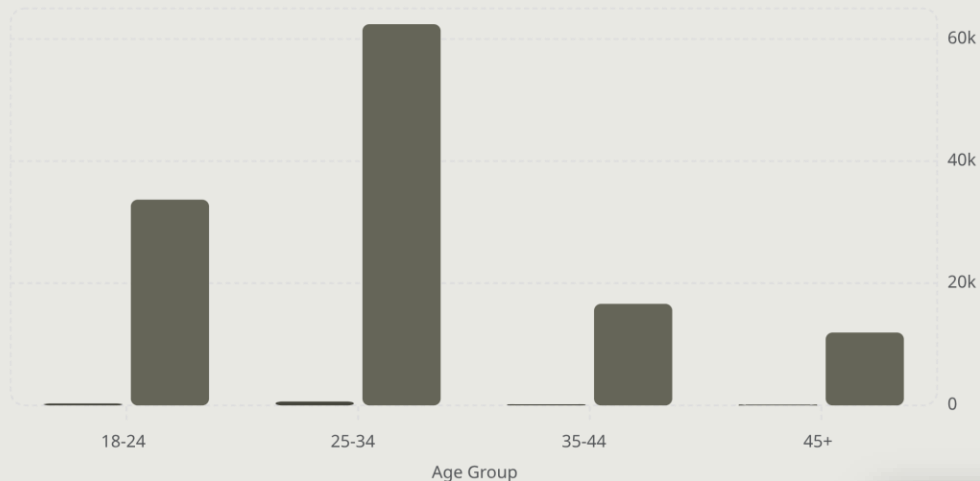
The 18-24 group placed **313 orders (25.04%)** with ₹33,685 in discounts – the second most engaged cohort, showing strong early-brand loyalty potential.

GENDER PARITY

Orders are nearly split: **Male 441, Female 433, and Other 376** – indicating broad, balanced appeal across genders.

■ Total Orders

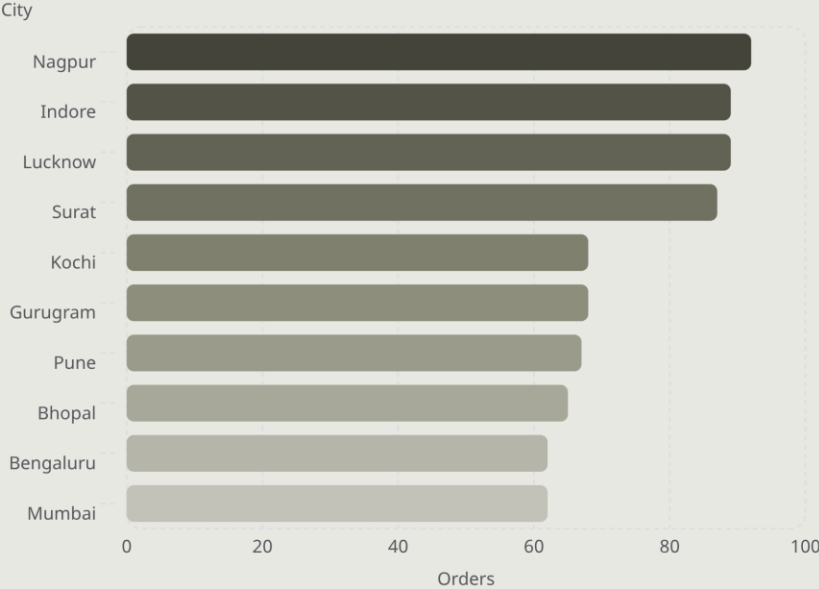
■ Total Discount (₹)



TOP CITIES & SALES CHANNELS

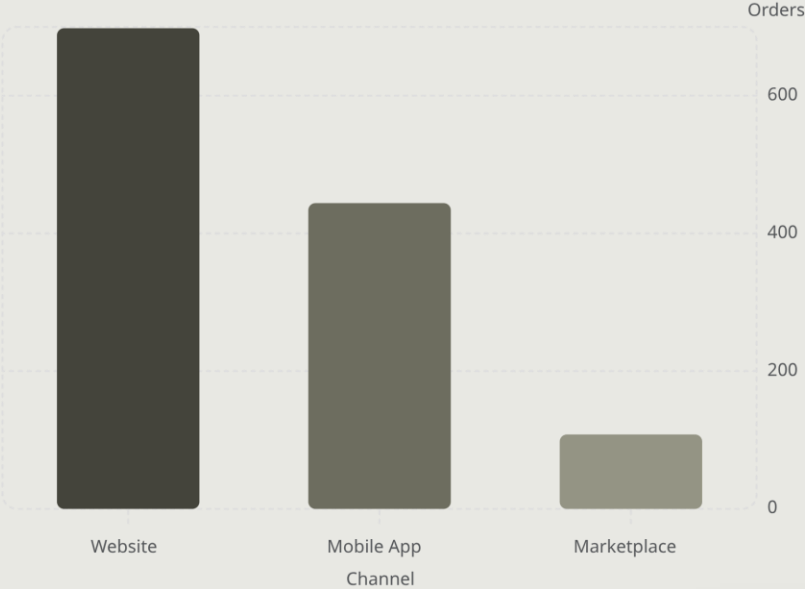
TOP 10 CITIES BY ORDERS

Nagpur leads with 92 orders, followed closely by **Indore** (89) and **Lucknow** (89). Tier-2 cities show strong purchasing activity, suggesting effective regional penetration.



SALES CHANNEL BREAKDOWN

The **Website** is the dominant channel with 698 orders, followed by **Mobile App** (444) and **Marketplace** (108). Digital-first strategy is clearly paying off.



TOP PRODUCTS BY RATING

Five products achieved a perfect **4.0 average rating**, reflecting strong customer satisfaction across skincare and haircare categories.



**AQUALOGICA
LIGHTWEIGHT
SUNSCREEN**

Avg Rating: **4.0**



**BODY LOTION
WITH
NIACINAMIDE**

Avg Rating: **4.0**



**BOND REPAIR
SHAMPOO**

Avg Rating: **4.0**



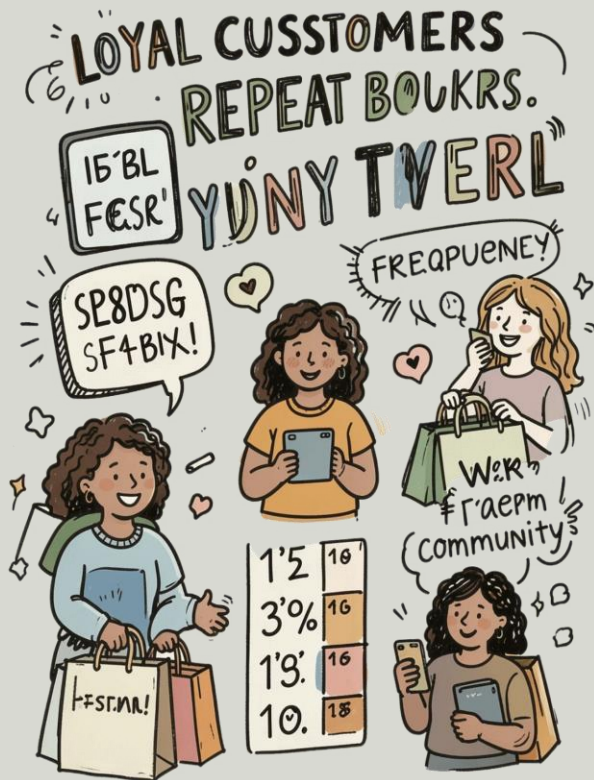
**GLYCOLIC
ACID FACE
CLEANSER**

Avg Rating: **4.0**



**HYALURONIC
ACID 2%
SERUM**

Avg Rating: **4.0**



REPEATED BUYERS & HIGH-SPENDING CUSTOMERS

Customers placing **more than 5 orders** were identified as repeated buyers. The top repeat buyer, **Aadhya Mishra**, placed 13 orders, alongside **Kiara Das** (13) and **Krishna Reddy** (12). A total of 20 customers met the repeat-buyer threshold, with order counts ranging from 9 to 13.

TOP REPEAT BUYERS

- Aadhya Mishra – 13 orders
- Kiara Das – 13 orders
- Krishna Reddy – 12 orders
- Meera Reddy – 12 orders
- Aditya Sharma – 12 orders

HIGH-SPENDING DISCOUNT USERS

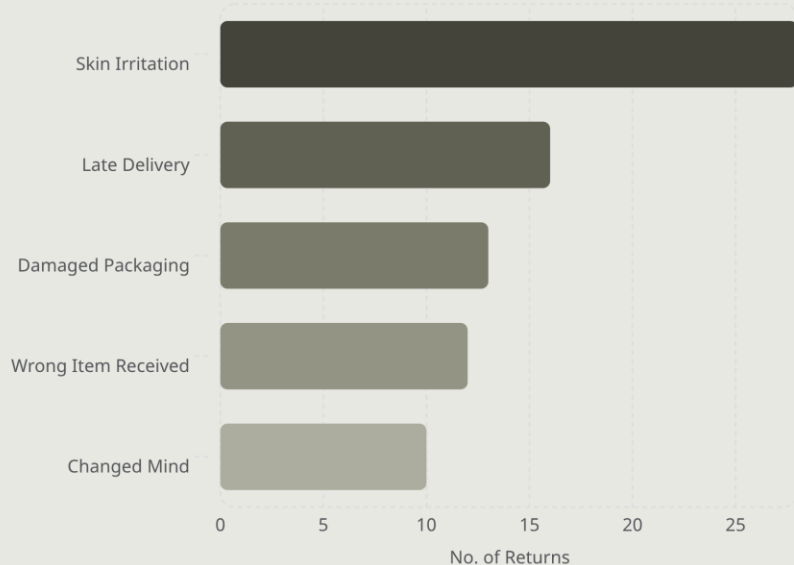
Customers who used discounts but still spent above the average purchase amount were identified. 100% clean. Top spenders include C0005 (₹2,047), C0020 (₹1,998), and C0013 (₹1,906).

RETURN REASONS & CUSTOMER ACQUISITION

TOP RETURN REASONS

Skin irritation is the leading cause of returns with 28 cases, followed by **late delivery** (16) and **damaged packaging** (13). Product formulation and logistics are key areas for improvement.

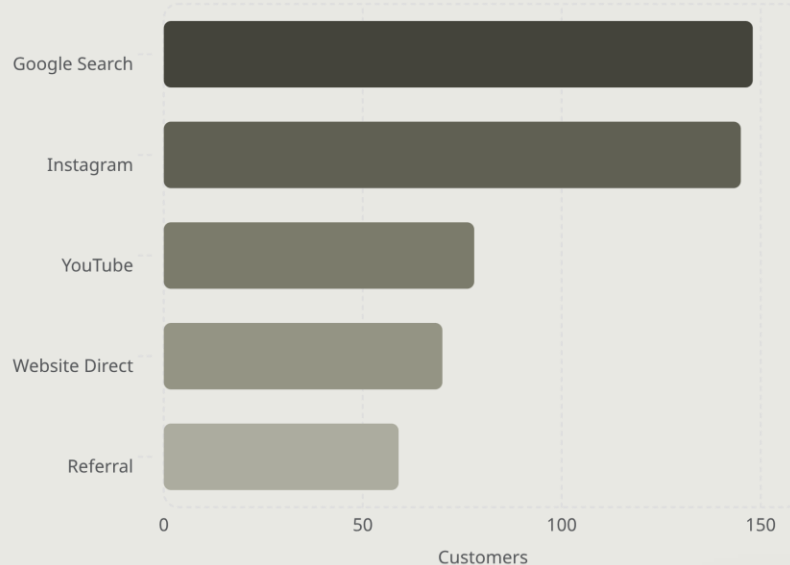
Return Reason



TOP ACQUISITION CHANNELS

Google Search (148 customers) and **Instagram** (145) are nearly tied as the top acquisition sources, together accounting for nearly 60% of all new customers. Paid search and social media are the primary growth engines.

Channel



POWER BI DASHBOARD HIGHLIGHTS

An interactive Power BI dashboard was built to present all insights visually, enabling stakeholders to explore data across customer behavior, gender, product category, and more.

500

NO. OF CUSTOMERS

2.04K

NO. OF ORDERS

₹1.12M

TOTAL FINAL AMOUNT

ORDERS BY AGE GROUP

25-34: 50.56% · 18-24: 25.04% · 35-44: 14.56% · 45+: 9.84%

ACQUISITION CHANNELS

Google Search: 29.6% · Instagram: 29% · Website Direct: 15.6% · Referral: 14%

TOP CITIES

Nagpur leads with the most orders, followed by Lucknow, Indore, Surat, and Kochi



KEY TAKEAWAYS

1 25-34 AGE GROUP IS THE CORE SEGMENT

Accounts for over half of all orders and the highest discount usage – prioritize this cohort in marketing and loyalty programs.

3 SKIN IRRITATION IS THE #1 RETURN DRIVER

Address product formulation and provide clearer ingredient guidance to reduce returns and improve satisfaction.

2 WEBSITE & MOBILE APP DRIVE SALES

Together they represent 91% of all orders. Invest in digital experience and app optimization for continued growth.

4 GOOGLE & INSTAGRAM ARE TOP ACQUISITION CHANNELS

Together they bring in nearly 60% of new customers – focus paid media budgets on search and social for maximum ROI.